Number Theory Makeup Midterm

50 minutes • 100 points • Six problems

Name: _____ **Date:** _____

Instructions. Answer all six problems and show your reasoning. No calculator is needed. You may quote results proved in class unless a proof is requested. Use the spaces provided; clearly label any work continued on the back. Reserve time to check your work.

1. Euclid and a linear congruence — 15 points

(a) Use the Euclidean algorithm to find $\gcd(195, 84)$. (5 points)

(b) Express this gcd as an integer linear combination of 195 and 84. (5 points)

(c) Find all solutions of $84x \equiv 6 \pmod{195}$. Give the distinct residues modulo 195 and explain why your list is complete. (5 points)

2. Continued fractions and an inverse — 15 points

Let $D(a) = \left(\begin{smallmatrix} a & 1 \\ 1 & 0 \end{smallmatrix}\right)$.

(a) Find the simple continued fraction of $139/60$ and all its convergents. Use a final digit at least 2. (6 points)

(b) Multiply the digit matrices for your expansion. Use the determinant of their product to find the inverse of 60 modulo 139. (9 points)

3. Reassembling residues and reducing powers **15 points**

(a) Solve $x \equiv 2 \pmod 5$ and $x \equiv 4 \pmod 7$. Justify uniqueness modulo 35. (7 points)

(b) Find the multiplicative order of 3 modulo 7 and use it to compute $3^{100} \pmod 7$. Show why your order is the least possible. (8 points)

4. Prime factors and divisor sums **15 points**

Here $\tau(n)$ counts positive divisors, $\varphi(n)$ counts units modulo n, and μ is the Möbius function.

(a) Factor 180 and compute $\tau(180)$, $\varphi(180)$, and $\mu(180)$. Show the prime-factor calculations. (6 points)

(b) Suppose f satisfies $\sum_{d \mid n} f(d) = n$ for every positive integer n. Use Möbius inversion to give a formula for $f(n)$ and compute $f(12)$. (9 points)

5. Valuations and a 5-adic limit **15 points**

For a prime p, let $v_p(r)$ denote the exponent of p in a nonzero rational r, with $v_p(a/b) = v_p(a) - v_p(b)$. Define $|r|_5 = 5^{-v_5(r)}$ for $r \neq 0$, $|0|_5 = 0$, and $d_5(x, y) = |x - y|_5$.

(a) State Legendre's formula for $v_p(n!)$ and use it to compute $v_3(30!)$. (5 points)

(b) Compute $|125/14|_5$ and $d_5(1/3, 16/3)$. (5 points)

(c) Let $S_n = \sum_{j=0}^{n} 5^j$. Find its 5-adic limit in $\mathbb{Q}$ and justify your answer by computing the distance to the limit. (5 points)

6. A field of formal pairs **25 points**

Use formal pairs $a + bs$ with $a, b \in \mathbb{F}_7$ and the relation $s^2 = 3$. Addition is coefficientwise; multiplication is distributive with s^2 replaced by 3. You may use the resulting commutative ring laws. All scalar arithmetic in this problem is modulo 7.

(a) List the squares in $\mathbb{F}_7$. Show that $a^2 - 3b^2 \neq 0$ whenever $(a, b) \neq (0, 0)$, and deduce that every nonzero pair has inverse $(a - bs)/(a^2 - 3b^2)$. (11 points)

(b) Compute $(2 + s)(3 + 2s)$ and $(1 + s)^{-1}$. Check the inverse by multiplication. (8 points)

(c) Give a basis over $\mathbb{F}_7$, the cardinality, and the characteristic. Explain why this field differs from $\mathbb{Z}/49\mathbb{Z}$. (6 points)

End of paper. Check that you have answered all six problems.